HRMS

11 簽核流程 — 系統設計書

目錄

HR11 WORKFLOW Service — System Design

系統名稱	版本
人力資源暨專案管理系統	**1.0**
日期	作者
2026-03-06	**HRMS 開發團隊**

簽核流程服務系統設計書

版本: 1.0 **日期:** 2025-12-07 **Domain代號:** 11 (WKF) **導入階段:** 第一階段(核心基礎服務) **目標:** 提供工程師完整的系統實作規格,供PM建立工項清單

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1. 服務概述

1.1 核心功能

- ✅ **流程定義:** 可視化流程設計
- ✅ **流程引擎:** 多級簽核、條件分流、平行會簽
- ✅ **代理人機制:** 請假/出差代理
- ✅ **逾時提醒:** 自動催辦

1.2 節點類型

節點類型	說明
START	開始節點
APPROVAL	審核節點 (單人/多人)
CONDITION	條件分流 (依金額/天數等)
PARALLEL	平行會簽 (需全部通過)
END	結束節點

2. UI設計

頁面代碼	頁面名稱	路由
`HR11-P01`	流程定義管理頁面	`/admin/workflows/definitions`
`HR11-P02`	流程設計器	`/admin/workflows/designer/:id`
`HR11-P03`	我的待辦頁面	`/profile/tasks`
`HR11-P04`	我的申請頁面	`/profile/applications`
`HR11-P05`	代理人設定頁面	`/profile/delegation`

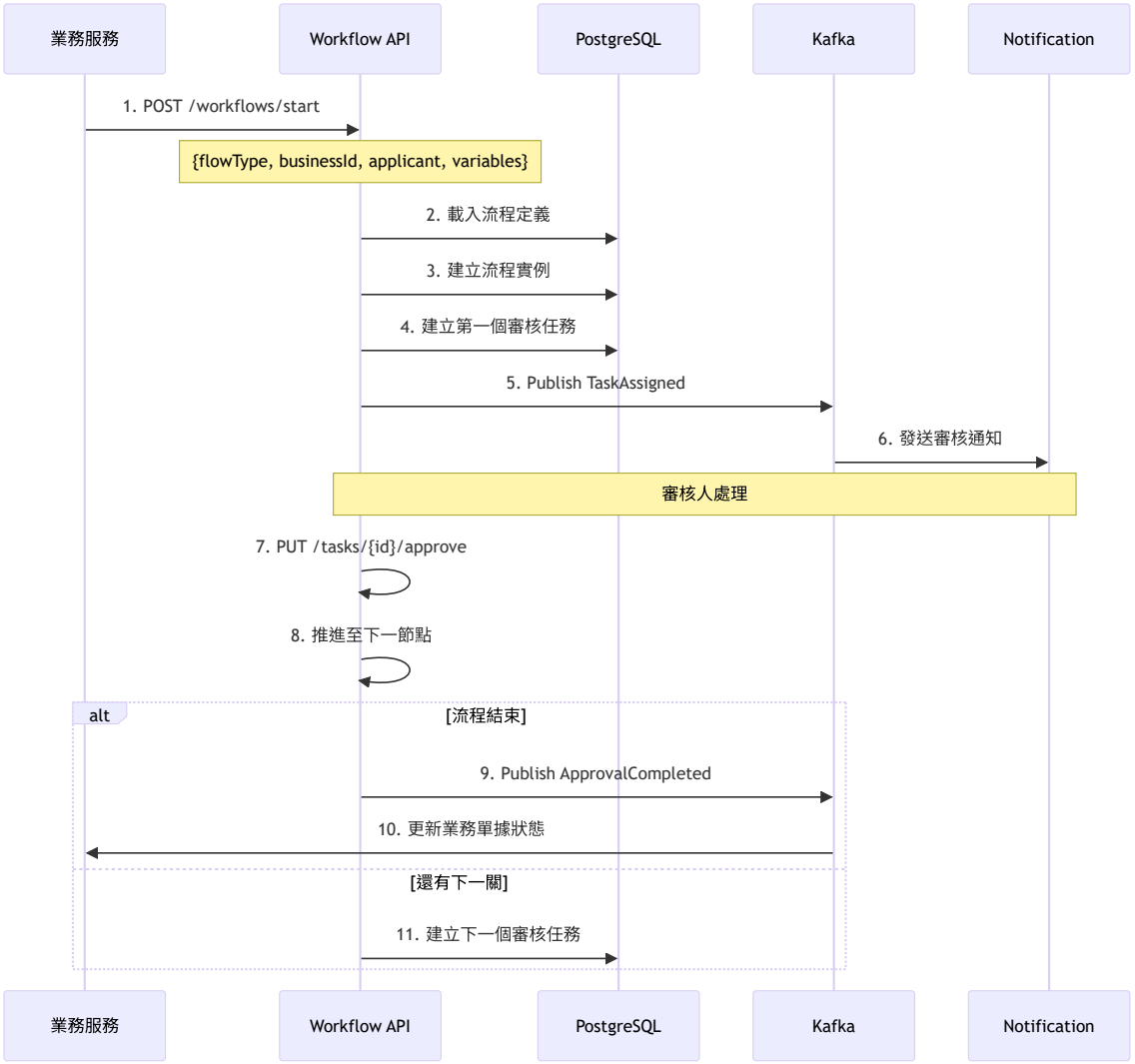
2.1 UI線稿

我的待辦頁面 (HR11-P03)

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                                 我的待辦頁面                                 │
│                                                                              │
│                              ┌──────────────────────────────┐                │
│                              │          我的待辦            │                │
│                              │──────────────────────────────│                │
│                              │                              │                │
│                              └──────────────────────────────┘                │
│                                              │                               │
│                                              ▼                               │
│                                                                              │
│                              待審核任務                                      │
│                              ┌──────────┬──────────┬──────┬──────────┬──────┬──────────┐
│                              │申請類型  │申請人    │內容  │提交時間  │逾期  │操作      │
│                              ├──────────┼──────────┼──────┼──────────┼──────┼──────────┤
│                              │請假申請  │張三      │特休2天│12/05     │-     │[✅][❌]  │
│                              │加班申請  │李四      │4小時 │12/06     │-     │[✅][❌]  │
│                              │請假申請  │王五      │病假1天│12/01     │⚠逾期 │[✅][❌]  │
│                              └──────────┴──────────┴──────┴──────────┴──────┴──────────┘
└──────────────────────────────────────────────────────────────────────────────┘
```

3. UX流程設計

3.1 審核流程執行



4. 畫面事件說明

4.1 我的待辦頁面事件 (HR11-P03)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-WKF-01`	頁面載入	onMount	載入待審核任務列表	GET /api/v1/workflows/tasks/pending
`E-WKF-02`	Tab切換	onChange	切換任務類型篩選	- (本地篩選)
`E-WKF-03`	核准按鈕	onClick	開啟核准確認對話框	-
`E-WKF-04`	核准確認	onClick	執行核准操作	PUT /api/v1/workflows/tasks/{id}/approve
`E-WKF-05`	駁回按鈕	onClick	開啟駁回原因對話框	-
`E-WKF-06`	駁回確認	onClick	執行駁回操作	PUT /api/v1/workflows/tasks/{id}/reject
`E-WKF-07`	查看詳情	onClick	開啟業務單據詳情Modal	-

E-WKF-04 詳細流程:

```
const handleApprove = async (taskId: string) => {
  try {
    // 1. 確認操作
    const confirmed = await Modal.confirm({
      title: '確認核准',
      content: '確定要核准此申請嗎？'
    });

    if (!confirmed) return;

    // 2. 呼叫API
    setLoading(true);
    const response = await workflowService.approveTask(taskId);

    // 3. 更新Redux狀態
    dispatch(removeTask(taskId));
    dispatch(refreshPendingCount());

    // 4. 顯示結果
    message.success('核准成功！');

    // 5. 重新載入待辦列表
    dispatch(loadPendingTasks());

  } catch (error) {
    message.error('核准失敗: ' + error.message);
  } finally {
    setLoading(false);
  }
};
```

4.2 我的申請頁面事件 (HR11-P04)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-APP-01`	頁面載入	onMount	載入我的申請列表	GET /api/v1/workflows/my/applications
`E-APP-02`	狀態篩選	onChange	篩選申請狀態	- (本地篩選)
`E-APP-03`	查看進度	onClick	開啟流程進度Modal	GET /api/v1/workflows/instances/{id}
`E-APP-04`	取消申請	onClick	取消審核中的申請	DELETE /api/v1/workflows/instances/{id}

4.3 代理人設定頁面事件 (HR11-P05)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-DEL-01`	頁面載入	onMount	載入代理人設定	GET /api/v1/workflows/delegations
`E-DEL-02`	新增代理人	onClick	開啟新增對話框	-
`E-DEL-03`	送出設定	onClick	建立代理人設定	POST /api/v1/workflows/delegations
`E-DEL-04`	刪除代理人	onClick	刪除代理人設定	DELETE /api/v1/workflows/delegations/{id}

4.4 流程設計器事件 (HR11-P02)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-DSN-01`	新增節點	onClick	在畫布上新增節點	-
`E-DSN-02`	連接節點	onConnect	建立節點間的連線	-
`E-DSN-03`	設定節點	onClick	開啟節點屬性設定	-
`E-DSN-04`	儲存流程	onClick	儲存流程定義	POST /api/v1/workflows/definitions
`E-DSN-05`	發布流程	onClick	發布流程為正式版本	PUT /api/v1/workflows/definitions/{id}/publish

5. Data Flow設計

5.1 前端狀態管理 (Redux)

5.1.1 State結構

```
interface WorkflowState {
  // 待辦任務
  tasks: {
    pendingTasks: ApprovalTask[];
    loading: boolean;
    totalCount: number;
  };

  // 我的申請
  myApplications: {
    applications: WorkflowInstance[];
    loading: boolean;
    filters: {
      status?: InstanceStatus;
      businessType?: string;
    };
  };

  // 代理人設定
  delegations: {
    list: Delegation[];
    loading: boolean;
  };

  // 流程定義（管理員）
  definitions: {
    list: WorkflowDefinition[];
    selectedDefinition: WorkflowDefinition | null;
    loading: boolean;
  };

  // 流程實例詳情
  instanceDetail: {
    instance: WorkflowInstance | null;
    tasks: ApprovalTask[];
    timeline: WorkflowTimeline[];
    loading: boolean;
  };
}

interface ApprovalTask {
  taskId: string;
  instanceId: string;
  nodeId: string;
```

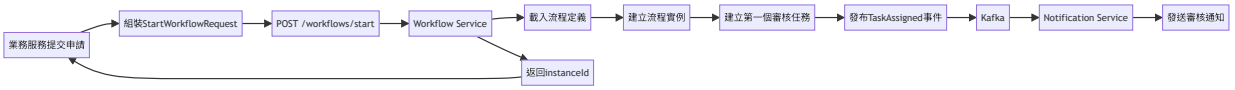
```
  businessType: string;
  businessId: string;
  applicant: {
    employeeId: string;
    fullName: string;
  };
  summary: string;
  assigneeId: string;
  status: TaskStatus;
  dueDate: string | null;
  isOverdue: boolean;
  createdAt: string;
}

interface WorkflowInstance {
  instanceId: string;
  definitionId: string;
  businessType: string;
  businessId: string;
  applicantId: string;
  currentNode: string;
  status: InstanceStatus;
  startedAt: string;
  completedAt: string | null;
}
```

5.1.2 Redux Actions

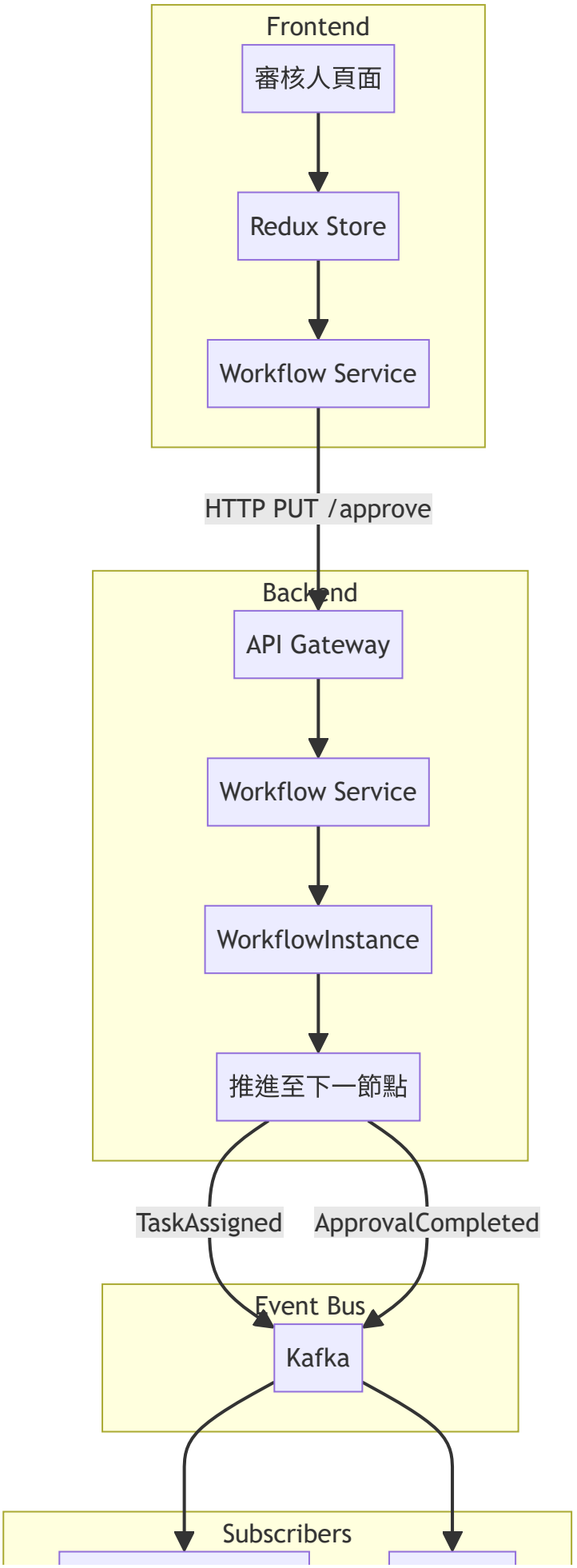
```
// 待辦任務Actions
export const taskActions = {
  loadPendingTasks: createAsyncThunk(
    'workflow/loadPendingTasks',
    async (employeeId: string) => {
      const response = await workflowService.getPendingTasks(employeeId);
      return response;
    }
  ),

  approveTask: createAsyncThunk(
    'workflow/approveTask',
    async (taskId: string) => {
      await workflowService.approveTask(taskId);
      return taskId;
    }
  ),

  rejectTask: createAsyncThunk(
    'workflow/rejectTask',
    async ({ taskId, reason }: { taskId: string; reason: string }) => {
      await workflowService.rejectTask(taskId, reason);
      return taskId;
    }
  ),
};

// 代理人Actions
export const delegationActions = {
  loadDelegations: createAsyncThunk(
    'workflow/loadDelegations',
    async (employeeId: string) => {
      const response = await workflowService.getDelegations(employeeId);
      return response;
    }
  ),

  createDelegation: createAsyncThunk(
    'workflow/createDelegation',
    async (request: CreateDelegationRequest) => {
      const response = await workflowService.createDelegation(request);
      return response;
    }
  ),
};
```

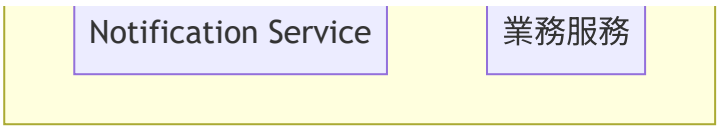
5.2 前後端資料流

5.2.1 業務服務啟動流程



5.2.2 審核流程推進


Frontend
審核人頁面
Redux Store
Workflow Service
HTTP PUT /approve
Backend
API Gateway
Workflow Service
WorkflowInstance
推進至下一節點
TaskAssigned
ApprovalCompleted
Event Bus
Kafka
Subscribers



5.3 業務整合範例

```
// Attendance Service 整合 Workflow
class CreateLeaveApplicationServiceImpl implements CommandApiService {
  @Autowired
  private WorkflowClient workflowClient;

  public CreateLeaveResponse execCommand(CreateLeaveRequest req) {
    // 1. 建立請假申請
    LeaveApplication leave = LeaveApplication.create(req);
    leaveRepository.save(leave);

    // 2. 啟動審核流程
    StartWorkflowRequest wfReq = new StartWorkflowRequest();
    wfReq.setFlowType("LEAVE_APPROVAL");
    wfReq.setBusinessId(leave.getId());
    wfReq.setApplicant(req.getEmployeeId());
    wfReq.setVariables(Map.of(
      "leaveType", leave.getLeaveType(),
      "totalDays", leave.getTotalDays()
    ));

    WorkflowInstance instance = workflowClient.startWorkflow(wfReq);

    // 3. 記錄workflow instance ID
    leave.setWorkflowInstanceId(instance.getInstanceId());
    leaveRepository.save(leave);

    return new CreateLeaveResponse(leave.getId(), instance.getInstanceId());
  }
}
```

6. 資料庫設計

```sql
-- 流程定義表
CREATE TABLE workflow_definitions (
    definition_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    flow_name VARCHAR(100) NOT NULL,
    flow_type VARCHAR(50) NOT NULL UNIQUE,
    nodes JSONB NOT NULL,
    edges JSONB NOT NULL,
    is_active BOOLEAN DEFAULT TRUE,
    version INTEGER DEFAULT 1,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 流程實例表
CREATE TABLE workflow_instances (
    instance_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    definition_id UUID NOT NULL REFERENCES workflow_definitions(definition_id),
    business_type VARCHAR(50) NOT NULL,
    business_id UUID NOT NULL,
    applicant_id UUID NOT NULL,
    current_node VARCHAR(100),
    status VARCHAR(20) DEFAULT 'RUNNING' CHECK (status IN ('RUNNING', 'COMPLETED', 'REJECTE
    started_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    completed_at TIMESTAMP,

    CONSTRAINT uk_instance UNIQUE (business_type, business_id)
);

CREATE INDEX idx_instance_status ON workflow_instances(status);
CREATE INDEX idx_instance_applicant ON workflow_instances(applicant_id);

-- 審核任務表
CREATE TABLE approval_tasks (
    task_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    instance_id UUID NOT NULL REFERENCES workflow_instances(instance_id),
    node_id VARCHAR(100) NOT NULL,
    assignee_id UUID NOT NULL,
    delegated_to UUID,
    status VARCHAR(20) DEFAULT 'PENDING' CHECK (status IN ('PENDING', 'APPROVED', 'REJECTED
    comments TEXT,
    due_date TIMESTAMP,
    is_overdue BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    completed_at TIMESTAMP
);

CREATE INDEX idx_task_assignee ON approval_tasks(assignee_id, status);
CREATE INDEX idx_task_instance ON approval_tasks(instance_id);
```

```sql
-- 代理人設定表
CREATE TABLE delegations (
    delegation_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    delegator_id UUID NOT NULL,
    delegatee_id UUID NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_delegation UNIQUE (delegator_id, start_date, end_date)
);
```

---

7. Domain設計

7.1 WorkflowInstance聚合根 (流程實例)

職責: 管理流程實例生命週期與狀態推進

Java實作:

```java
@Entity
@Table(name = "workflow_instances")
public class WorkflowInstance {
    @EmbeddedId
    private InstanceId id;
    private UUID definitionId;
    private String businessType;
    private UUID businessId;
    private UUID applicantId;
    private String currentNode;

    @Enumerated(EnumType.STRING)
    private InstanceStatus status;

    /**
     * 推進到下一節點
     */
    public void advanceToNode(String nextNode) {
        this.currentNode = nextNode;

        if ("END".equals(nextNode)) {
            this.status = InstanceStatus.COMPLETED;
            DomainEventPublisher.publish(new ApprovalCompletedEvent(
                this.businessType,
                this.businessId
            ));
        }
    }

    /**
     * 駁回
     */
    public void reject(String reason) {
        this.status = InstanceStatus.REJECTED;
        DomainEventPublisher.publish(new ApprovalRejectedEvent(
            this.businessType,
            this.businessId,
            reason
        ));
    }
}
```

7.2 ApprovalTask聚合根 (審核任務)

職責: 管理審核任務的指派與處理

Java實作:

```java
@Entity
@Table(name = "approval_tasks")
public class ApprovalTask {
    @EmbeddedId
    private TaskId id;

    @Column(name = "instance_id", nullable = false)
    private UUID instanceId;

    @Column(name = "node_id", nullable = false)
    private String nodeId;

    @Column(name = "assignee_id", nullable = false)
    private UUID assigneeId;

    @Column(name = "delegated_to")
    private UUID delegatedTo;

    @Enumerated(EnumType.STRING)
    @Column(name = "status", nullable = false)
    private TaskStatus status;

    @Column(name = "comments")
    private String comments;

    @Column(name = "due_date")
    private LocalDateTime dueDate;

    @Column(name = "is_overdue")
    private boolean isOverdue;

    // ========== Factory Method ==========

    /**
     * 建立審核任務
     */
    public static ApprovalTask create(
            UUID instanceId,
            String nodeId,
            UUID assigneeId,
            Integer dueDays) {

        ApprovalTask task = new ApprovalTask();
        task.id = TaskId.generate();
        task.instanceId = instanceId;
        task.nodeId = nodeId;
        task.assigneeId = assigneeId;
        task.status = TaskStatus.PENDING;

        if (dueDays != null && dueDays > 0) {
            task.dueDate = LocalDateTime.now().plusDays(dueDays);
```

```
        }

        // 發布任務指派事件
        DomainEventPublisher.publish(new TaskAssignedEvent(
            task.id.getValue(),
            task.instanceId,
            task.assigneeId
        ));

        return task;
    }

    /**
     * 核准任務
     */
    public void approve(UUID approverId, String comments) {
        if (this.status != TaskStatus.PENDING) {
            throw new DomainException("只能核准待處理的任務");
        }

        this.status = TaskStatus.APPROVED;
        this.comments = comments;

        DomainEventPublisher.publish(new TaskApprovedEvent(
            this.id.getValue(),
            this.instanceId,
            approverId
        ));
    }

    /**
     * 駁回任務
     */
    public void reject(UUID approverId, String reason) {
        if (this.status != TaskStatus.PENDING) {
            throw new DomainException("只能駁回待處理的任務");
        }

        Objects.requireNonNull(reason, "駁回必須提供原因");

        this.status = TaskStatus.REJECTED;
        this.comments = reason;

        DomainEventPublisher.publish(new TaskRejectedEvent(
            this.id.getValue(),
            this.instanceId,
            approverId,
            reason
        ));
    }
```

```java
    /**
     * 檢查並標記逾期
     */
    public void checkOverdue() {
        if (this.dueDate != null &&
            LocalDateTime.now().isAfter(this.dueDate) &&
            this.status == TaskStatus.PENDING) {

            this.isOverdue = true;

            DomainEventPublisher.publish(new TaskOverdueEvent(
                this.id.getValue(),
                this.instanceId,
                this.assigneeId
            ));
        }
    }
}
```

7.3 Delegation實體 (代理人設定)

```java
@Entity
@Table(name = "delegations")
public class Delegation {
    @EmbeddedId
    private DelegationId id;

    @Column(name = "delegator_id", nullable = false)
    private UUID delegatorId;

    @Column(name = "delegatee_id", nullable = false)
    private UUID delegateeId;

    @Column(name = "start_date", nullable = false)
    private LocalDate startDate;

    @Column(name = "end_date", nullable = false)
    private LocalDate endDate;

    @Column(name = "is_active")
    private boolean isActive;

    /**
     * 檢查是否在生效期間
     */
    public boolean isEffective(LocalDate date) {
        return isActive &&
               !date.isBefore(startDate) &&
               !date.isAfter(endDate);
    }

    /**
     * 檢查並停用過期的代理設定
     */
    public void checkExpiry() {
        if (LocalDate.now().isAfter(this.endDate)) {
            this.isActive = false;
        }
    }
}
```

8. 領域事件設計

8.1 事件總覽

事件名稱	觸發時機	訂閱服務	重要性
`TaskAssignedEvent`	任務指派給審核人	Notification	⭐⭐⭐
`TaskApprovedEvent`	任務核准	Workflow內部	⭐⭐
`TaskRejectedEvent`	任務駁回	Workflow內部	⭐⭐
`ApprovalCompletedEvent`	流程核准完成	業務服務 (Attendance等)	⭐⭐⭐⭐⭐
`ApprovalRejectedEvent`	流程駁回	Notification, 業務服務	⭐⭐⭐⭐
`TaskOverdueEvent`	任務逾期	Notification	⭐⭐

8.2 事件Schema

8.2.1 TaskAssignedEvent

```json
{
  "eventId": "evt-wkf-001",
  "eventType": "TaskAssigned",
  "timestamp": "2025-12-06T10:00:00Z",
  "payload": {
    "taskId": "550e8400-e29b-41d4-a716-446655440000",
    "instanceId": "550e8400-e29b-41d4-a716-446655440001",
    "businessType": "LEAVE_APPLICATION",
    "businessId": "app-leave-001",
    "applicant": {
      "employeeId": "emp-001",
      "fullName": "張三"
    },
    "assignee": {
      "employeeId": "mgr-001",
      "fullName": "李經理"
    },
    "taskSummary": "員工張三申請特休假2天",
    "dueDate": "2025-12-08T10:00:00Z",
    "businessUrl": "/attendance/leave/applications/app-leave-001"
  }
}
```

訂閱者處理:

- **Notification Service:** 發送審核通知給李經理 (Email + 系統內通知)

8.2.2 ApprovalCompletedEvent (關鍵事件)

```
{
  "eventId": "evt-wkf-002",
  "eventType": "ApprovalCompleted",
  "timestamp": "2025-12-06T15:30:00Z",
  "payload": {
    "instanceId": "550e8400-e29b-41d4-a716-446655440001",
    "businessType": "LEAVE_APPLICATION",
    "businessId": "app-leave-001",
    "applicantId": "emp-001",
    "finalApprover": {
      "employeeId": "mgr-002",
      "fullName": "王總監"
    },
    "approvalPath": [
      {
        "nodeId": "node-1",
        "approver": "李經理",
        "approvedAt": "2025-12-06T11:00:00Z"
      },
      {
        "nodeId": "node-2",
        "approver": "王總監",
        "approvedAt": "2025-12-06T15:30:00Z"
      }
    ],
    "totalDuration": "5.5h"
  }
}
```

訂閱者處理:

- **Attendance Service:** 將請假申請狀態更新為 `APPROVED`
- **Notification Service:** 通知申請人審核通過

8.2.3 ApprovalRejectedEvent

```
{
  "eventId": "evt-wkf-003",
  "eventType": "ApprovalRejected",
  "timestamp": "2025-12-06T12:00:00Z",
  "payload": {
    "instanceId": "550e8400-e29b-41d4-a716-446655440002",
    "businessType": "OVERTIME_APPLICATION",
    "businessId": "ot-002",
    "applicantId": "emp-003",
    "rejectedBy": {
      "employeeId": "mgr-001",
      "fullName": "李經理"
    },
    "rejectedAt": "2025-12-06T12:00:00Z",
    "rejectionReason": "加班時數已達本月上限",
    "nodeId": "node-1"
  }
}
```

訂閱者處理:

- Overtime Service: 將加班申請狀態更新為 `REJECTED`
- Notification Service: 通知申請人駁回原因

8.2.4 TaskOverdueEvent

```
{
  "eventId": "evt-wkf-004",
  "eventType": "TaskOverdue",
  "timestamp": "2025-12-08T10:00:00Z",
  "payload": {
    "taskId": "550e8400-e29b-41d4-a716-446655440003",
    "instanceId": "550e8400-e29b-41d4-a716-446655440004",
    "businessType": "LEAVE_APPLICATION",
    "businessId": "app-leave-005",
    "assignee": {
      "employeeId": "mgr-001",
      "fullName": "李經理"
    },
    "dueDate": "2025-12-08T10:00:00Z",
    "overdueHours": 2,
    "applicant": {
      "employeeId": "emp-005",
      "fullName": "趙六"
    }
  }
}
```

訂閱者處理:

- **Notification Service:** 發送逾期提醒給審核人及其主管

---

9. API設計 (10個端點)

端點	方法	Controller
`/api/v1/workflows/definitions`	POST	HR11DefinitionCmdController
`/api/v1/workflows/definitions`	GET	HR11DefinitionQryController
`/api/v1/workflows/start`	POST	HR11InstanceCmdController
`/api/v1/workflows/instances/{id}`	GET	HR11InstanceQryController
`/api/v1/workflows/tasks/pending`	GET	HR11TaskQryController
`/api/v1/workflows/tasks/{id}/approve`	PUT	HR11TaskCmdController
`/api/v1/workflows/tasks/{id}/reject`	PUT	HR11TaskCmdController
`/api/v1/workflows/my/applications`	GET	HR11InstanceQryController
`/api/v1/workflows/delegations`	POST	HR11DelegationCmdController
`/api/v1/workflows/delegations`	GET	HR11DelegationQryController

---

10. 工項清單摘要

10.1 前端開發工項

工項編號	工項名稱	說明	預估工時 (人天)
FE-WKF-01	HR11-P01 流程定義管理頁面	流程列表、啟用/停用	2
FE-WKF-02	HR11-P02 流程設計器	可視化流程設計器 (React Flow)	8
FE-WKF-03	HR11-P03 我的待辦頁面	待審核任務列表、核准/駁回	3
FE-WKF-04	HR11-P04 我的申請頁面	申請進度追蹤	2
FE-WKF-05	HR11-P05 代理人設定頁面	代理人CRUD	2
FE-WKF-06	流程進度Modal	顯示審核時間軸	1
FE-WKF-07	Redux狀態管理	Workflow State、Actions	2
FE-WKF-08	API Service層	WorkflowService、API封裝	1
FE-WKF-09	Factory層	ViewModel轉換	1
FE-WKF-10	單元測試	Factory、Component測試	2
小計			**24人天**

10.2 後端開發工項

工項編號	工項名稱	說明	預估工時 (人天)
BE-WKF-01	WorkflowInstance聚合根	流程實例Domain邏輯	3
BE-WKF-02	ApprovalTask聚合根	審核任務Domain邏輯	3
BE-WKF-03	Delegation實體	代理人設定	1
BE-WKF-04	Repository層	Instance、Task、Delegation Repository	2
BE-WKF-05	流程定義管理API	建立/查詢/發布流程定義 (2端點)	2
BE-WKF-06	流程啟動API	啟動流程、查詢實例 (2端點)	3
BE-WKF-07	任務審核API	核准/駁回/查詢待辦 (3端點)	3
BE-WKF-08	代理人API	代理人CRUD (2端點)	1
BE-WKF-09	我的申請API	查詢我的申請 (1端點)	1
BE-WKF-10	流程引擎Service	節點推進、條件分流、平行會簽邏輯	5
BE-WKF-11	事件發布	6個領域事件	2
BE-WKF-12	逾期檢查Job	Quartz定時任務	1
BE-WKF-13	單元測試	Domain層100%覆蓋率	3
BE-WKF-14	整合測試	API端點測試	2
小計			**32人天**

10.3 測試工項

工項編號	工項名稱	說明	預估工時 (人天)
TEST-WKF-01	Domain單元測試	WorkflowInstance、ApprovalTask	2
TEST-WKF-02	API整合測試	10個端點	2
TEST-WKF-03	流程引擎測試	多級簽核、條件分流、平行會簽	3
TEST-WKF-04	事件發布測試	6個事件Schema驗證	1
TEST-WKF-05	E2E測試	完整審核流程	2
小計			**10人天**

10.4 文件工項

工項編號	工項名稱	預估工時 (人天)
DOC-WKF-01	API文件 (Swagger)	1
DOC-WKF-02	流程設計器使用手冊	1
DOC-WKF-03	業務整合指南	1
小計		**3人天**

10.5 總工時估計

類別	工時 (人天)
前端開發	24
後端開發	32
測試	10
文件	3
總計	**69人天**

備註:

- 流程設計器為技術難點,建議使用成熟的第三方庫 (React Flow、bpmn-js)
- 流程引擎需支援動態載入流程定義,避免硬編碼
- 建議分兩階段實作:
  - **階段一:** 基礎審核流程 (單人審核、多級審核) - 40人天
  - **階段二:** 進階功能 (條件分流、平行會簽、可視化設計器) - 29人天

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